Supplementary Table 1: Correspondence between parameters named in the manuscript and variables in the associated code (publicly available on GitHub).

Description	Paper	Code
Recorded density	$D_{p,t}$	dens
SD of density at the log-scale	σ_{obs}	sigma_obs
Natural mortality rate	m	est_m
Fishing mortality rate	$f_{a,t}$	f
Intercept for detection probability	β_0	beta_obs_int
Coefficient linking detection probability to density	β_1	beta_obs
Overall average recruitment	μ	mean_recruits
Conditional variance in the AR1 process	σ_{proc}	sigma_r
White noise used in the AR1 process	z_t	raw
Autcorrelation from AR1 term	α	alpha
Excess in natural mortality due to temperature	γ	m_e
Isotropic dispersal rate	δ	d
Sea bottom temperature	$T_{p,t}$	sbt
How much tax per unit of temperature	β_{tax}	beta_t
Suitability index	$I_{p,t}$	T_adjust
Temperature optimizing $I_{p,t}$	τ	Topt
Width parameter from $I_{p,t}$	ω	width

Supplementary Table 2: Hyperparameters used in the DRM.

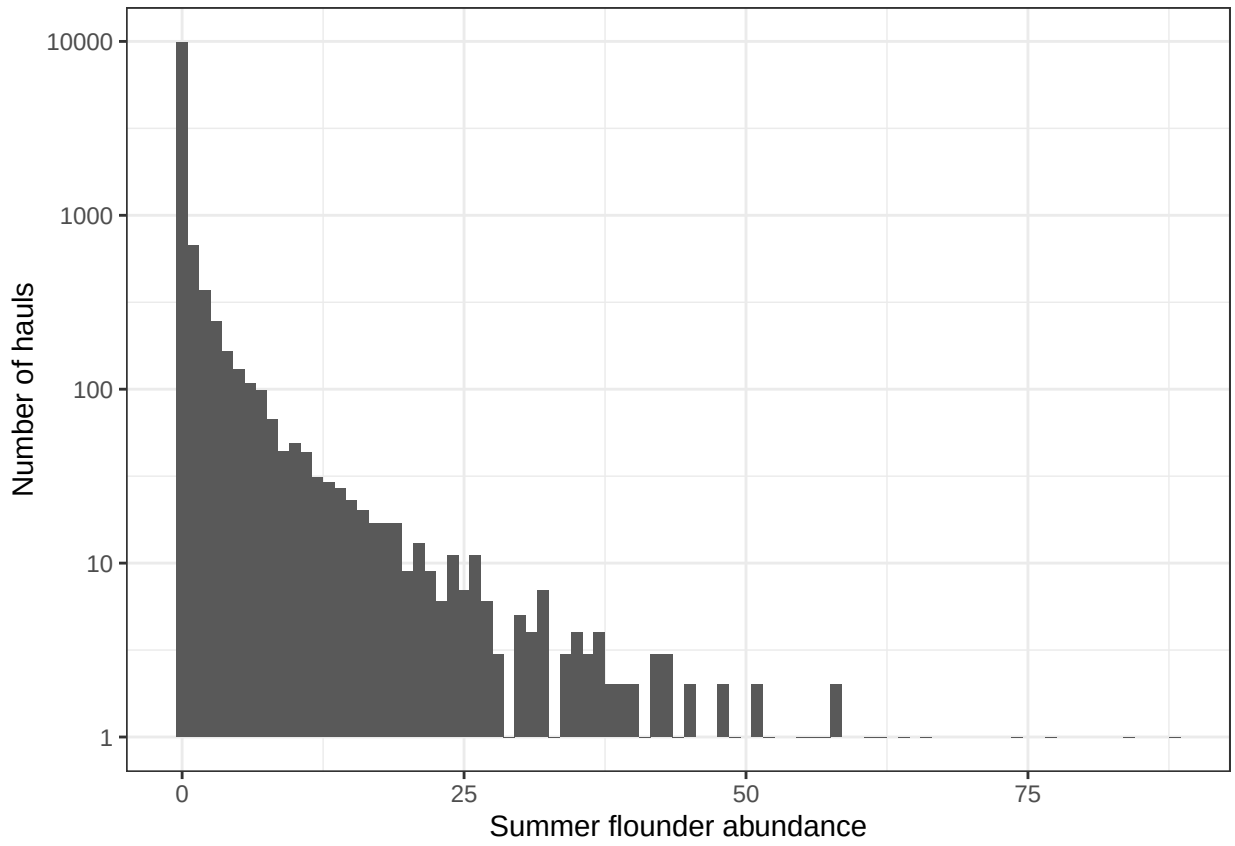
Parameter	Prior.Distribution
$\beta_{obs,1}$	Normal $\sim (0.001, 0.1)$
$\beta_{obs,0}$	Normal $\sim (-100, 4)$
z	Normal $\sim (0, 1)$
σ_{proc}	Normal $\sim (0.2, 0.1)$
σ_{obs}	Normal $\sim (0.2, 0.2)$
$width$	Normal $\sim (4, 2)$
T_{opt}	Normal $\sim (18, 2)$
d	Normal $\sim (0.01, 0.1)$
β_{tax}	Normal $\sim (0, 2)$
α	Beta $\sim (12, 20)$
μ_r	Lognormal $\sim (7, 5)$
m	Normal $\sim (0.25, 0.1)$
γ	Exponential ~ 2.3

Supplementary Table 3: Bayesian linear regressions of sea bottom temperature (measured in the trawl surveys) on year within each patch during the model training interval (1972-2006) with Gaussian errors, summarized by the median and 95% Highest Posterior Density Interval (HPDI) of the posterior estimate for the slope term. Values are truncated at three significant digits.

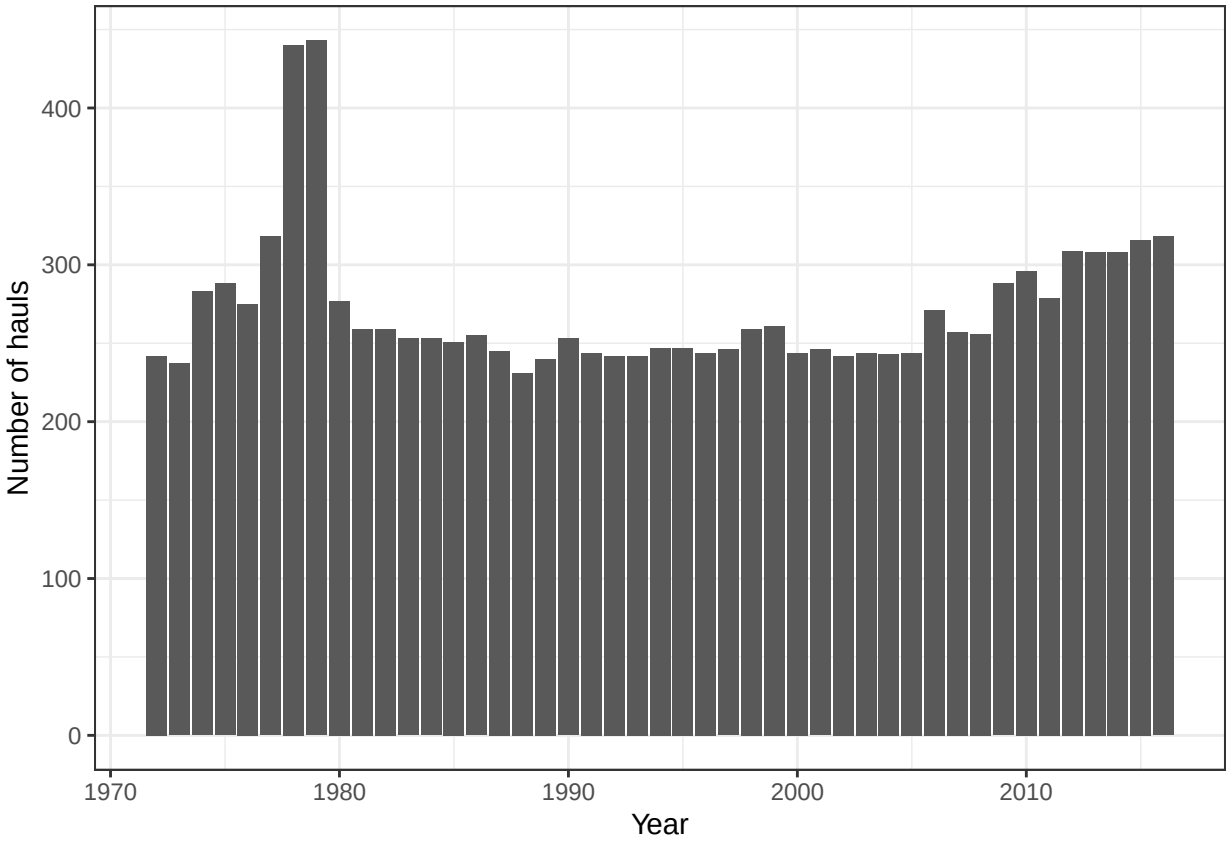
Patch	Median	HPDI
35	0.00236	-0.0734 — 0.0803
36	0.04540	0.00128 — 0.0909
37	0.05610	0.0106 — 0.0972
38	0.02900	-0.00582 — 0.0654
39	0.04710	0.0203 — 0.0709
40	0.01940	0.00837 — 0.0309
41	0.02680	0.0108 — 0.0428
42	-0.00264	-0.0103 — 0.00479
43	0.00328	-0.00566 — 0.0123
44	0.00607	-0.0154 — 0.0253

Supplementary Table 4: Bayesian linear regressions of sea bottom temperature (measured in the trawl surveys) on year within each patch during the model testing interval (2007-2016) with Gaussian errors, summarized by the median and 95% Highest Posterior Density Interval (HPDI) of the posterior estimate for the slope term. Values are truncated at three significant digits.

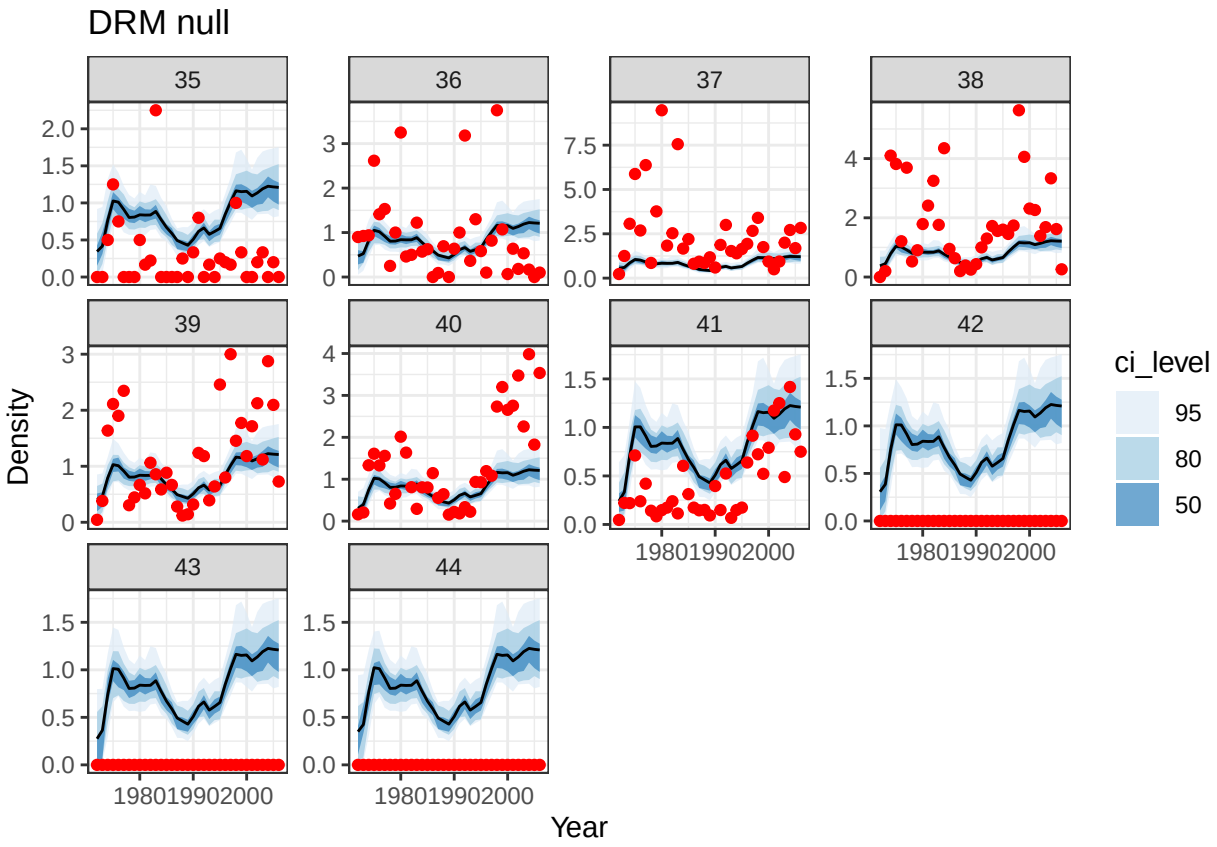
Patch	Median	HPDI
35	0.1520	-0.31 — 0.598
36	0.2310	-0.139 — 0.561
37	0.1620	-0.134 — 0.477
38	-0.0439	-0.266 — 0.17
39	0.0890	-0.0928 — 0.261
40	0.3610	0.276 — 0.446
41	0.1590	0.0441 — 0.278
42	0.1690	0.114 — 0.228
43	0.1740	0.106 — 0.243
44	0.2530	0.159 — 0.342



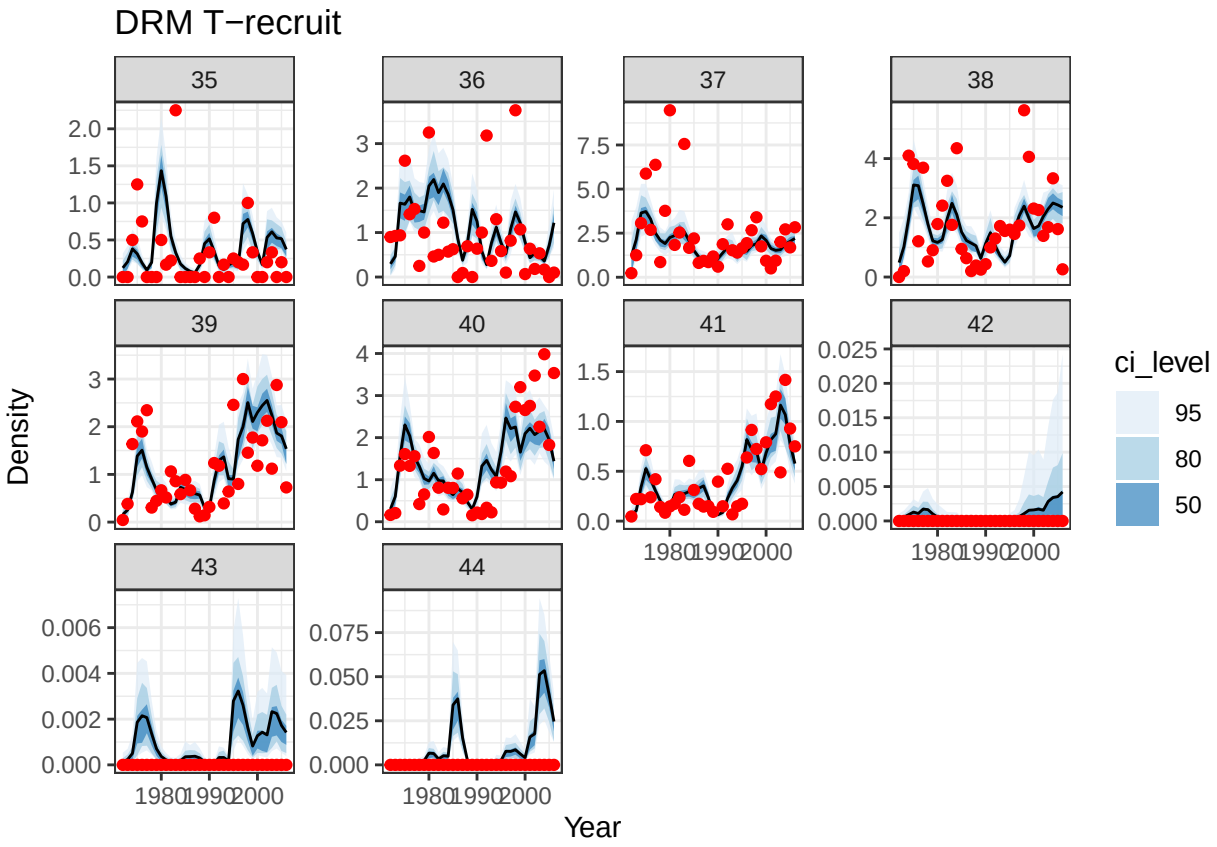
Supplementary Figure 1: Frequency distribution of summer flounder abundance in all hauls used in the analysis (testing and training data combined; n = 12,203).



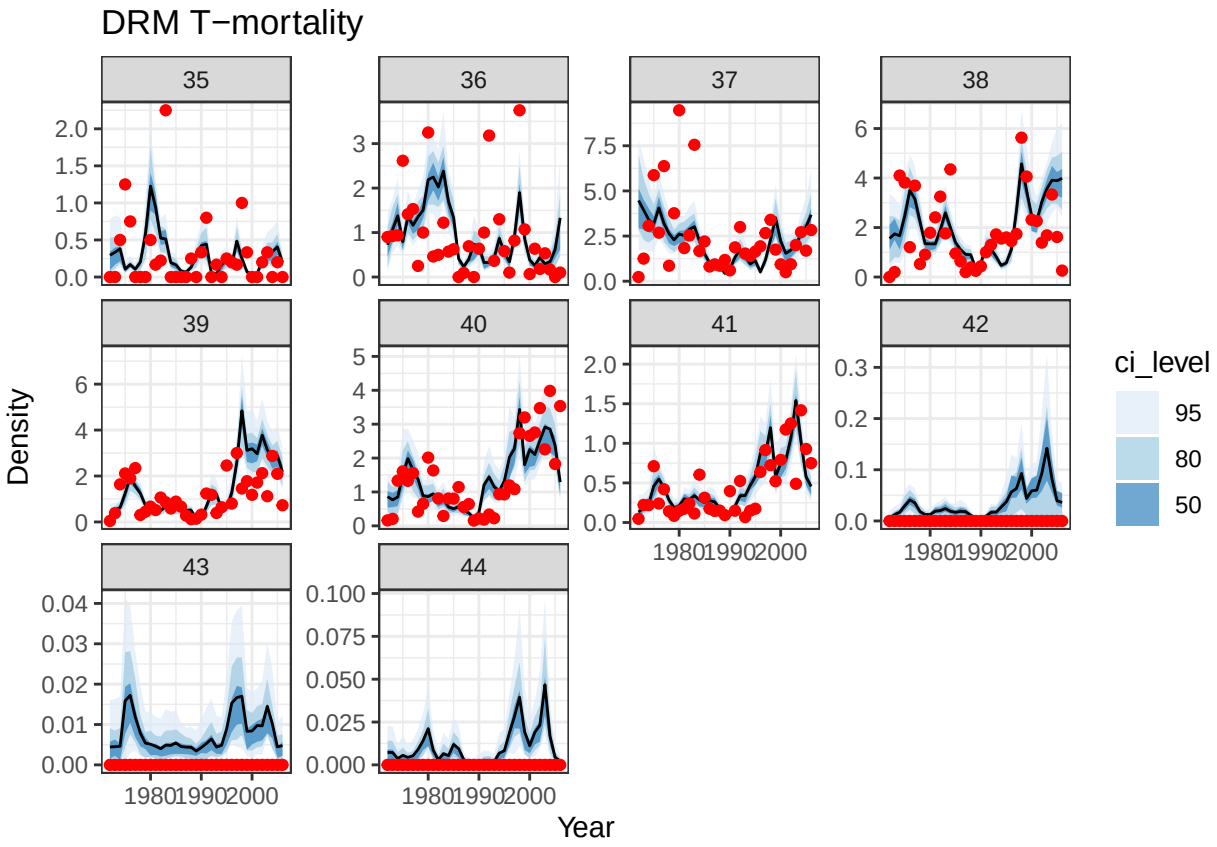
Supplementary Figure 2: Number of hauls per year; n = 12,203.



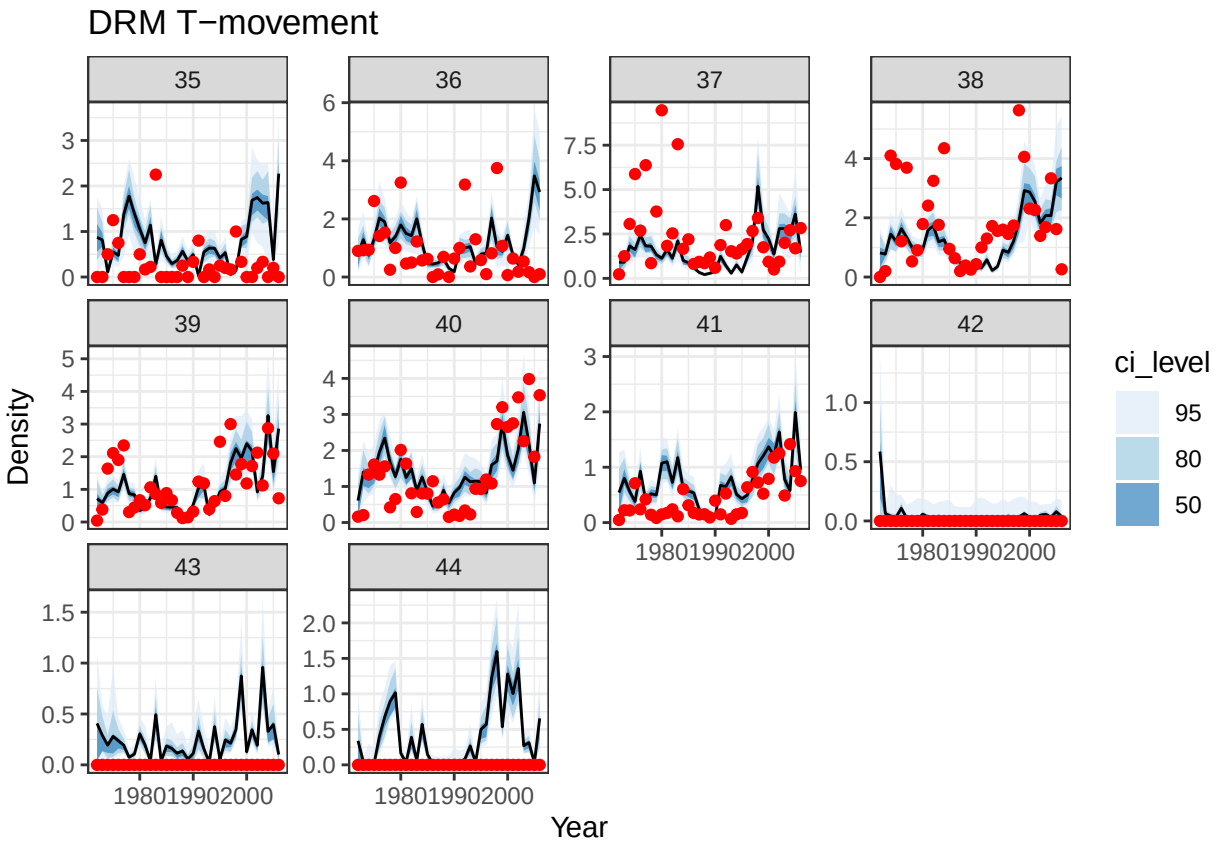
Supplementary Figure 3: Median density estimated by the null DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



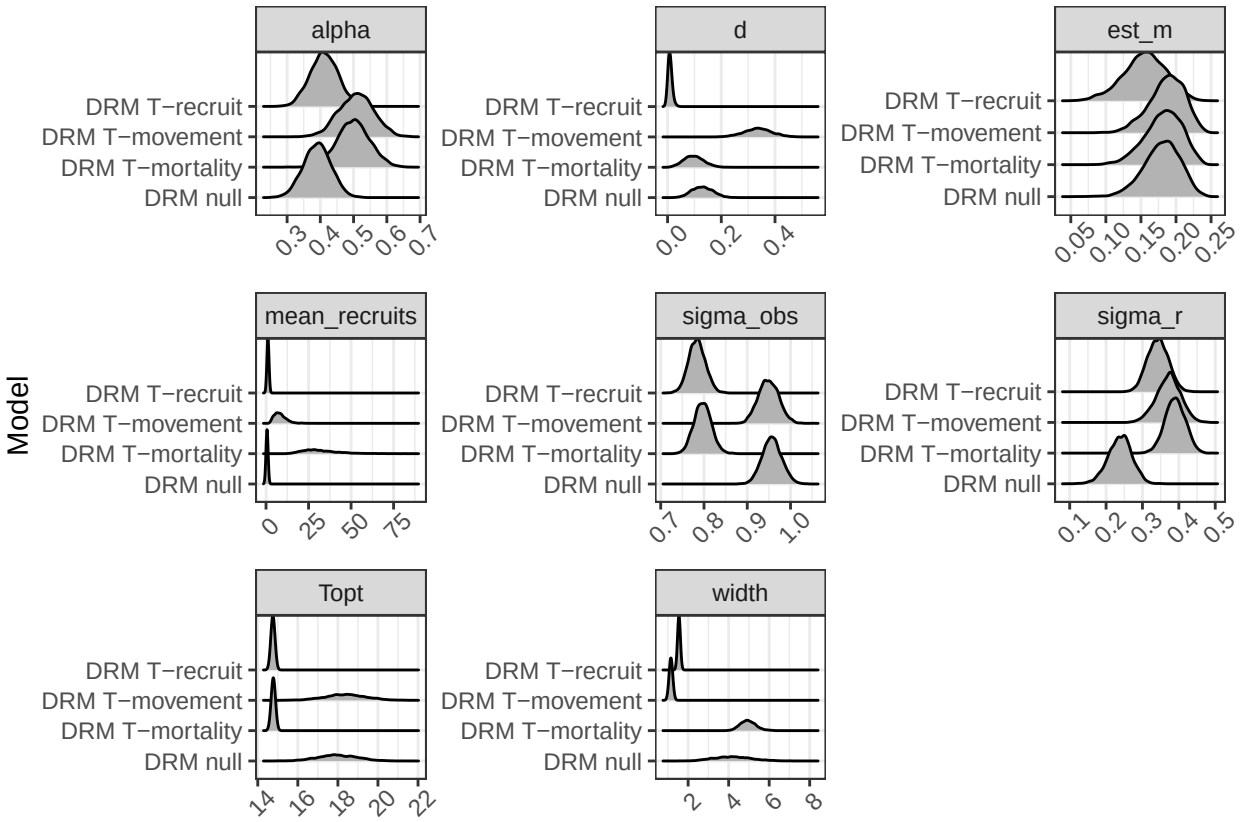
Supplementary Figure 4: Median density estimated by the temperature-dependent recruitment DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



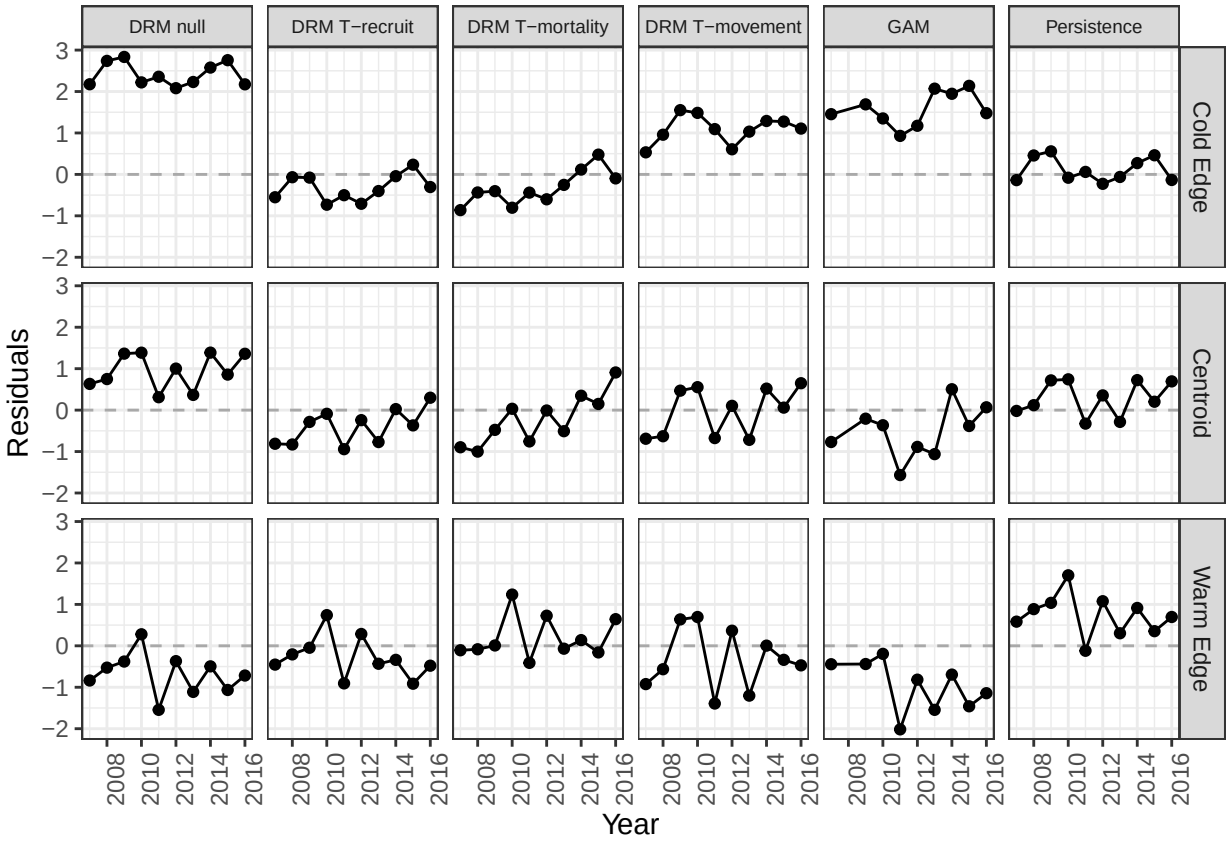
Supplementary Figure 5: Median density estimated by the temperature-dependent mortality DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



Supplementary Figure 6: Median density estimated by the temperature-dependent movement DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



Supplementary Figure 7: Posterior distributions of parameters from the four fitted DRMs. These parameters are d, the annual dispersal fraction between adjacent patches; mean_recruits, the average density of recruits per patch; sigma_obs, the standard deviation of density at the log scale; Topt, the optimal temperature estimated for whichever process (movement or recruitment or mortality) was estimated as temperature-dependent, in °C; and width, a parameter controlling how sensitive the modeled process was to temperature. See Supp. Tab. 1 for correspondence of code variables to model parameters in the manuscript.



Supplementary Figure 8: Residuals (estimated minus observed values) of range metrics each year over the ten-year testing period for the focal models in the main text: DRM with no temperature effect (DRM null) or a temperature effect on recruitment, mortality, or movement; a GAM SDM; and a persistence forecast. The horizontal dashed line indicates zero residuals, i.e., the estimate perfectly matches the testing data.

We used R v. 4.4.0 (R Core Team 2024*a*) and the following R packages: bookdown v. 0.42 (Xie 2016, 2025), brms v. 2.22.0 (Bürkner 2017, 2018, 2021), broom.mixed v. 0.2.9.7 (Bolker and Robinson 2026), cmdstanr v. 0.8.1 (Gabry et al. 2024), coda v. 0.19.4.1 (Plummer et al. 2006), data.table v. 1.16.4 (Barrett et al. 2024), directlabels v. 2024.1.21 (Hocking 2024), doParallel v. 1.0.17 (Corporation and Weston 2022), foreach v. 1.5.2 (Microsoft and Weston 2022), ggdist v. 3.3.2 (Kay 2024*a*, 2024*b*), ggh4x v. 0.3.0 (van den Brand 2024), ggrepel v. 0.9.6 (Slowikowski 2024), ggridges v. 0.5.6 (Wilke 2024), ggspatial v. 1.1.9 (Dunnington 2023), grid v. 4.4.0 (R Core Team 2024*b*), here v. 1.0.1 (Müller 2020), kableExtra v. 1.4.0 (Zhu 2024), knitr v. 1.49 (Xie 2014, 2015, 2024), Matrix v. 1.7.0 (Bates et al. 2024), mgcv v. 1.9.1 (Wood 2003, 2004, 2011; Wood et al. 2016; Wood 2017), parallel v. 4.4.0 (R Core Team 2024*c*), patchwork v. 1.3.0 (Pedersen 2024), posterior v. 1.6.0 (Vehtari et al. 2021; Bürkner et al. 2024), rmarkdown v. 2.29 (Xie et al. 2018, 2020; Allaire et al. 2024), rnaturalearth v. 1.0.1 (Massicotte and South 2023), rnaturalearthdata v. 1.0.0 (South et al. 2024), rstan v. 2.32.6 (Stan Development Team 2024), sf v. 1.0.19 (Pebesma 2018; Pebesma and Bivand 2023), tidybayes v. 3.0.7 (Kay 2024*c*), tidytext v. 0.4.2.9000 (Silge and Robinson 2016), tidyverse v. 2.0.0 (Wickham et al. 2019).

References

Allaire, J., Y. Xie, C. Dervieux, J. McPherson, J. Luraschi, K. Ushey, A. Atkins, et al. 2024. rmarkdown: Dynamic documents for r.

Barrett, T., M. Dowle, A. Srinivasan, J. Gorecki, M. Chirico, T. Hocking, and B. Schwendinger. 2024. data.table: Extension of "data.frame".

Bates, D., M. Maechler, and M. Jagan. 2024. Matrix: Sparse and dense matrix classes and methods.

Bolker, B., and D. Robinson. 2026. broom.mixed: Tidying methods for mixed models.

Bürkner, P.-C. 2017. brms: An R package for Bayesian multilevel models using Stan. Journal of Statistical Software 80:1–28.

———. 2018. Advanced Bayesian multilevel modeling with the R package brms. The R Journal 10:395–411.

———. 2021. Bayesian item response modeling in R with brms and Stan. Journal of Statistical Software 100:1–54.

Bürkner, P.-C., J. Gabry, M. Kay, and A. Vehtari. 2024. posterior: Tools for working with posterior distributions.

Corporation, M., and S. Weston. 2022. doParallel: Foreach parallel adaptor for the "parallel" package.

Dunnington, D. 2023. ggspatial: Spatial data framework for ggplot2.

Gabry, J., R. Češnovar, A. Johnson, and S. Bronder. 2024. cmdstanr: R interface to "CmdStan".

Hocking, T. D. 2024. directlabels: Direct labels for multicolor plots.

Kay, M. 2024*a*. ggdist: Visualizations of distributions and uncertainty in the grammar of graphics. IEEE Transactions on Visualization and Computer Graphics 30:414–424.

———. 2024*b*. ggdist: Visualizations of distributions and uncertainty.

———. 2024*c*. tidybayes: Tidy data and geoms for Bayesian models.

Massicotte, P., and A. South. 2023. rnaturalearth: World map data from natural earth.

Microsoft, and S. Weston. 2022. foreach: Provides foreach looping construct.

Müller, K. 2020. here: A simpler way to find your files.

Pebesma, E. 2018. Simple Features for R: Standardized Support for Spatial Vector Data. The R Journal 10:439–446.

Pebesma, E., and R. Bivand. 2023. Spatial Data Science: With applications in R. Chapman and Hall/CRC.

Pedersen, T. L. 2024. patchwork: The composer of plots.

Plummer, M., N. Best, K. Cowles, and K. Vines. 2006. CODA: Convergence diagnosis and output analysis for MCMC. R News 6:7–11.

R Core Team. 2024*a*. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

———. 2024*b*. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

———. 2024*c*. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

Silge, J., and D. Robinson. 2016. tidytext: Text mining and analysis using tidy data principles in r. JOSS 1.

Slowikowski, K. 2024. ggrepel: Automatically position non-overlapping text labels with "ggplot2".

South, A., S. Michael, and P. Massicotte. 2024. rnaturalearthdata: World vector map data from natural earth used in "rnaturalearth".

Stan Development Team. 2024. RStan: The R interface to Stan.

van den Brand, T. 2024. ggh4x: Hacks for "ggplot2".

Vehtari, A., A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner. 2021. Rank-normalization, folding, and localization: An improved rhat for assessing convergence of MCMC (with discussion). Bayesian Analysis.

Wickham, H., M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Grolemund, et al. 2019. Welcome to the tidyverse. Journal of Open Source Software 4:1686.

Wilke, C. O. 2024. ggridges: Ridgeline plots in "ggplot2".

Wood, S. N. 2003. Thin-plate regression splines. Journal of the Royal Statistical Society (B) 65:95–114.

———. 2004. Stable and efficient multiple smoothing parameter estimation for generalized additive models. Journal of the American Statistical Association 99:673–686.

———. 2011. Fast stable restricted maximum likelihood and marginal likelihood estimation of semipara-

metric generalized linear models. Journal of the Royal Statistical Society (B) 73:3–36.
Wood, S. N. 2017. Generalized additive models: An introduction with r (2nd ed.). Chapman; Hall/CRC.
Wood, S. N., N., Pya, and B. S"afken. 2016. Smoothing parameter and model selection for general smooth models (with discussion). Journal of the American Statistical Association 111:1548–1575.
Xie, Y. 2014. knitr: A comprehensive tool for reproducible research in R. *in* V. Stodden, F. Leisch, and R. D. Peng, eds. Implementing reproducible computational research. Chapman; Hall/CRC.
———. 2015. Dynamic documents with R and knitr (2nd ed.). Chapman; Hall/CRC, Boca Raton, Florida.
———. 2016. bookdown: Authoring books and technical documents with R markdown. Chapman; Hall/CRC, Boca Raton, Florida.
———. 2024. knitr: A general-purpose package for dynamic report generation in r.
———. 2025. bookdown: Authoring books and technical documents with r markdown.
Xie, Y., J. J. Allaire, and G. Grolemund. 2018. R markdown: The definitive guide. Chapman; Hall/CRC, Boca Raton, Florida.
Xie, Y., C. Dervieux, and E. Riederer. 2020. R markdown cookbook. Chapman; Hall/CRC, Boca Raton, Florida.
Zhu, H. 2024. kableExtra: Construct complex table with “kable” and pipe syntax.